MICROPROCESSOR & COMPUTER ARCHITECTURE

8086 / 8088 Microprocessors

Architecture, Buses & Processor Evolution

80286 · 80386 · Pentium Chips

Complete Study Guide for B.Tech | Diploma | Competitive Examinations

13 Chapters	Exam Q&A	Comparison Tables	ASCII Diagrams

Syllabus Coverage: Introduction to 8086 | Introduction to 8088 | Architecture of 8086/8088 | Base, Data, Address & Control Bus | Introduction to 80286 | Introduction to 80386 | Introduction to Pentium Chips

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CHAPTER 1

Introduction to Microprocessor

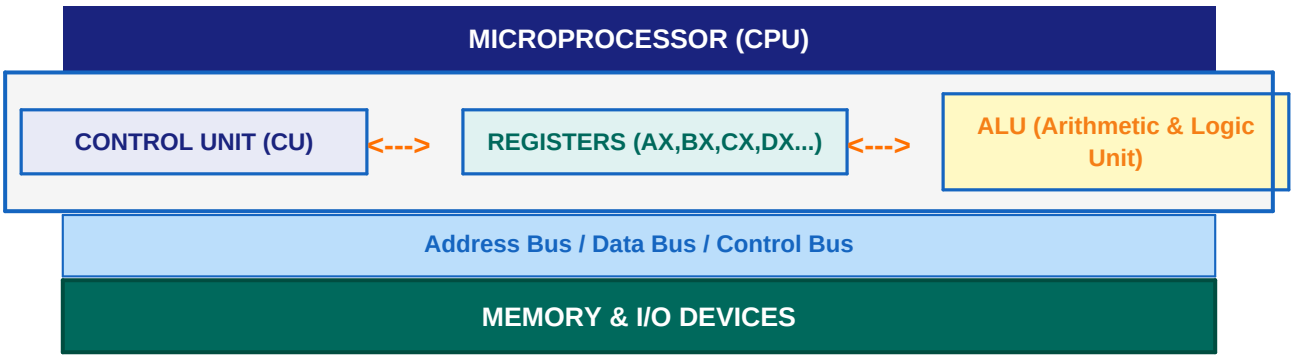
1.1 Definition of Microprocessor

Definition: A **microprocessor** is a single Integrated Circuit (IC) chip that contains the complete functional units of a Central Processing Unit (CPU). It can perform arithmetic operations, logical operations, data transfer, and control operations by fetching instructions from memory, decoding them, and executing them.

Key Point: Microprocessor = ALU + Control Unit + Registers — all on a single chip.
It is the **brain of a computer** — it processes all data and controls all operations.

1.2 Basic Components of a Microprocessor

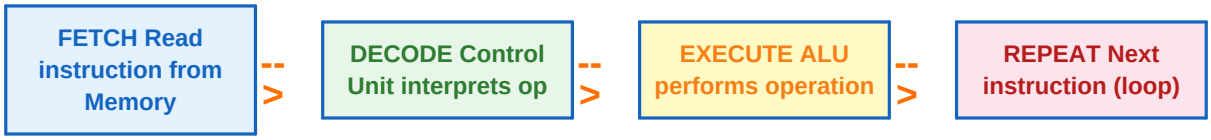
Unit	Full Name	Role
ALU	Arithmetic & Logic Unit	Performs arithmetic (+, –, ×, ÷) and logic (AND, OR, NOT, XOR) operations
CU	Control Unit	Fetches, decodes instructions and generates control signals for execution
Registers	High-Speed Memory inside CPU	Temporarily holds data, addresses and results during processing



1.3 How a Microprocessor Works

A microprocessor works in a repeated **Fetch** → **Decode** → **Execute** cycle:

- **Fetch:** Read the next instruction from memory using the Address Bus
- **Decode:** Control Unit interprets the instruction to know what to do
- **Execute:** ALU performs the operation; result is stored in register or memory



1.4 Types of Microprocessors

Type	Examples	Application
4-bit	Intel 4004	Calculators, simple toys
8-bit	Intel 8085, Z80	Early computers, embedded systems
16-bit	Intel 8086, 8088	Personal computers (IBM PC era)
32-bit	Intel 80386, Pentium	Modern PCs and servers (older)
64-bit	Intel Core i-series, Ryzen	Modern computers, smartphones

1.5 Applications of Microprocessors

- **Personal Computers** – Desktops, laptops (Intel, AMD processors)
- **Embedded Systems** – Washing machines, microwave ovens, AC controllers
- **Mobile Devices** – Smartphones and tablets (ARM-based processors)
- **Industrial Automation** – CNC machines, robotic arms, PLCs
- **Medical Equipment** – ECG machines, ventilators, blood pressure monitors
- **Communication** – Routers, modems, satellite systems

1.6 Advantages and Disadvantages

Advantages	Disadvantages
Very small size — single IC chip	Generates heat at high speeds
High processing speed	Needs external RAM, ROM, I/O chips
Low cost due to mass production	Complex to design and program
Low power consumption	Limited by word size (8/16/32/64-bit)
Highly reliable — no moving parts	Sensitive to power supply fluctuations

CHAPTER 2

Introduction to 8086 Microprocessor

2.1 Definition of 8086 Microprocessor

Definition: The **Intel 8086** is a **16-bit microprocessor** introduced by Intel Corporation in **1978**. It is fabricated using **HMOS technology** and is housed in a **40-pin DIP (Dual In-line Package)**. The 8086 has a **16-bit data bus** and a **20-bit address bus**, enabling it to address up to **1 MB (1,048,576 bytes)** of memory. It was the **first processor in the x86 family**, which forms the foundation of most modern PC architectures. The 8086 operates at clock frequencies of **5 MHz, 8 MHz, and 10 MHz** and contains approximately **29,000 transistors**.

In simple terms, the 8086 is a programmable chip that acts as the **brain of a computer**. It fetches instructions stored in memory, decodes (interprets) them, and executes them to produce a result. All arithmetic, logic, data transfer, and control operations of a computer system are managed by the 8086.

Property	Value / Description
Full Name	Intel 8086 Microprocessor
Manufacturer	Intel Corporation
Year of Introduction	1978
Architecture	16-bit (CISC – Complex Instruction Set Computer)
Data Bus Width	16 bits (D0–D15)
Address Bus Width	20 bits (A0–A19)
Maximum Addressable Memory	1 MB = 2²■ bytes = 1,048,576 bytes
Clock Speed	5 MHz / 8 MHz / 10 MHz
Number of Pins	40 pins (DIP package)
Technology	HMOS (High-performance MOS)
Number of Transistors	~29,000
Instruction Set	x86 (16-bit)
Operating Modes	Minimum Mode and Maximum Mode
Predecessor	Intel 8085 (8-bit)
Successor	Intel 8088 (1979), Intel 80186 (1982)

2.2 What is a Microprocessor?

A **microprocessor** is a single integrated circuit (IC) chip that contains the functions of a central processing unit (CPU). It is the heart of a computer system, capable of fetching instructions from memory, decoding

them, and executing them. It processes data by performing arithmetic, logic, and control operations.

Simple Definition: A microprocessor = ALU + Control Unit + Registers, all on a single chip.

2.3 Evolution of Microprocessors

Generation	Processor	Year	Bits	Key Feature
1st	Intel 4004	1971	4-bit	First commercial microprocessor
2nd	Intel 8008	1972	8-bit	8-bit data processing
3rd	Intel 8085	1976	8-bit	Multiplexed address/data bus
4th	Intel 8086	1978	16-bit	First x86 processor, 1 MB memory
5th	Intel 80286	1982	16-bit	Protected mode introduced
6th	Intel 80386	1985	32-bit	Full 32-bit processor
7th	Pentium	1993	32-bit	Superscalar, dual pipeline

2.4 Features of 8086 Microprocessor

- 16-bit microprocessor – processes 16 bits of data at a time
- 20-bit address bus – can address up to 2^{20} = 1,048,576 bytes = **1 MB** of memory
- 16-bit data bus – transfers 16 bits of data in one bus cycle
- Clock speed: **5 MHz, 8 MHz, or 10 MHz** depending on variant (8086, 8086-2, 8086-1)
- Contains **14 registers** of 16-bit size each
- Supports **pipelined architecture** with a 6-byte instruction queue (prefetch queue)
- Can operate in two modes: **Minimum Mode** and **Maximum Mode**
- Two functional units: **Bus Interface Unit (BIU)** and **Execution Unit (EU)**
- Memory divided into **four segments**: Code, Data, Stack, Extra
- Supports both **memory-mapped I/O** and **I/O-mapped I/O**
- Can handle **256 interrupt types**
- Manufactured using **HMOS technology** (High-performance Metal-Oxide Semiconductor)
- Available in **40-pin DIP** (Dual In-line Package)

2.5 Technical Specifications of 8086

Parameter	Specification
Data Bus Width	16 bits
Address Bus Width	20 bits
Addressable Memory	1 MB (1,048,576 bytes)
Clock Frequency	5 MHz / 8 MHz / 10 MHz
Instruction Queue	6 bytes
Number of Registers	14 registers (all 16-bit)
Technology	HMOS

Supply Voltage	+5V
Package	40-pin DIP
Instruction Set	x86 (16-bit)
Operating Modes	Minimum Mode / Maximum Mode
Max. I/O Ports	65,536 (64K)
Transistors	~29,000

2.6 Functional Overview

The 8086 works in a pipelined fashion. While the **Execution Unit (EU)** is executing an instruction, the **Bus Interface Unit (BIU)** simultaneously fetches the next instruction from memory and loads it into the instruction queue. This overlap of fetch and execute cycles increases throughput and makes the processor more efficient.

2.7 Applications of 8086

- Personal computers and workstations (IBM PC and IBM PC/XT used 8088, compatible with 8086)
- Embedded systems and industrial controllers
- Scientific and engineering calculations
- Communication systems and modems
- Educational and experimental microprocessor kits
- Data acquisition and control systems

2.8 Advantages and Limitations of 8086

Advantages	Limitations
16-bit processing – faster than 8-bit processors	Only 1 MB addressable memory
Pipelined architecture improves performance	No built-in floating-point unit (needs 8087 co-processor)
Large 1 MB memory space for its era	Complex segmented memory model
Separate I/O address space (64K ports)	Limited to 16-bit operations
Strong industry support and software base	Obsolete for modern applications

CHAPTER 3

Introduction to 8088 Microprocessor

3.1 Overview of 8088

The **Intel 8088** was introduced in **1979**, one year after the 8086. It is essentially a cost-reduced version of the 8086. Internally, it is identical to the 8086 (16-bit internal architecture), but its **external data bus is only 8 bits wide**. This made it cheaper to interface with existing 8-bit peripheral chips.

The 8088 gained enormous popularity when IBM chose it for the **original IBM PC (1981)**, making it one of the most significant microprocessors in computing history.

> **Key Insight:** 8088 = 8086 with an 8-bit external data bus. All internal registers and operations remain 16-bit. Memory addressing is still 20-bit (1 MB).

3.2 Features and Specifications of 8088

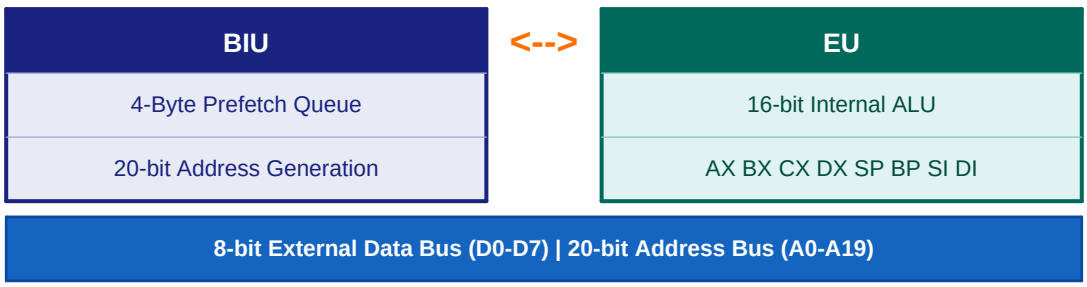
- **Internal data bus: 16-bit** (same as 8086)
- **External data bus: 8-bit** (different from 8086)
- **Address bus: 20-bit** – same 1 MB addressable memory as 8086
- Clock speed: **5 MHz or 8 MHz**
- **4-byte instruction prefetch queue** (8086 has 6-byte queue)
- Same **14 internal registers** as 8086
- Same **Minimum/Maximum mode** operation as 8086
- Same **instruction set** as 8086 (fully compatible)
- Manufactured in **HMOS technology**
- Available in **40-pin DIP** package
- Uses **two bus cycles** to transfer 16-bit data (since bus is 8-bit)
- Compatible with Intel 8085 peripheral support chips

3.3 Internal Organization of 8088

The internal organization of the 8088 is identical to the 8086. It has two main units:

1. Bus Interface Unit (BIU): Fetches instructions and data from memory, writes data to memory, and manages the external bus. The BIU in 8088 works with an 8-bit external bus but performs internal 16-bit operations.

2. Execution Unit (EU): Decodes and executes instructions received from the BIU. The EU always works with 16-bit data internally.



3.4 Applications of 8088

- IBM PC (1981) – the most famous application of 8088
- IBM PC/XT (1983) – extended version using 8088
- Portable and low-cost personal computers of early 1980s
- Embedded control applications
- Educational and hobbyist computer systems

3.5 Advantages and Limitations of 8088

- **Advantages:** Lower cost than 8086 (cheaper bus interface), fully compatible with 8086 software, compatible with 8-bit peripheral chips, chosen for IBM PC which created a massive software ecosystem
- **Limitations:** Slower than 8086 for 16-bit data transfers (needs two 8-bit cycles), 8-bit external bus is a bottleneck, same memory limitation as 8086 (1 MB)

CHAPTER 4

Comparison of 8086 and 8088

4.1 Key Architectural Differences

Although the 8086 and 8088 share the same internal architecture and instruction set, they differ in their external bus interface, which affects performance and cost.

4.2 Detailed Comparison Table

Feature	8086	8088
Introduction Year	1978	1979
Internal Data Bus	16-bit	16-bit
External Data Bus	16-bit	8-bit
Address Bus	20-bit	20-bit
Max. Addressable Memory	1 MB	1 MB
Instruction Prefetch Queue	6 bytes	4 bytes
Clock Speed	5 / 8 / 10 MHz	5 / 8 MHz
Bus Cycles for 16-bit Transfer	1 cycle	2 cycles
Pin Count	40 pins (DIP)	40 pins (DIP)
Technology	HMOS	HMOS
Peripheral Compatibility	16-bit peripherals	8-bit peripherals
Cost	Higher	Lower
Famous Application	Used in embedded systems	IBM PC (1981)
Performance	Faster (16-bit bus)	Slightly slower
Instruction Set	x86 16-bit (identical)	x86 16-bit (identical)
Operating Modes	Min / Max Mode	Min / Max Mode

4.3 Data Bus Difference – Impact on Performance

The most critical difference is the **external data bus width**. When the 8086 transfers a 16-bit (2-byte) value, it does so in a single bus cycle. The 8088 must perform **two consecutive 8-bit bus cycles** to transfer the same 16-bit value. This makes the 8088 approximately **10-15% slower** than the 8086 for programs that frequently use 16-bit data.

8086 – 16-bit transfer in 1 bus cycle

Byte 1 + Byte 2 (simultaneous on D0-D15)

8088 – 16-bit transfer needs 2 bus cycles

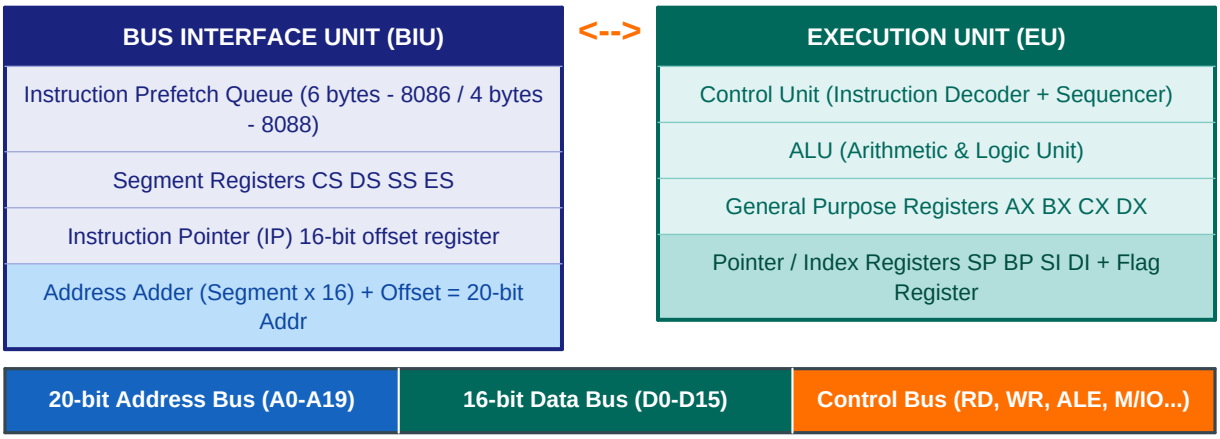
Byte 1 (D0-D7) – Cycle 1	Byte 2 (D0-D7) – Cycle 2

CHAPTER 5

Architecture of 8086/8088 Microprocessors

5.1 Block Diagram Overview

The 8086/8088 microprocessor is divided internally into two independent functional units that operate simultaneously (in parallel), giving a pipelined effect:



5.2 Bus Interface Unit (BIU)

The BIU is responsible for all bus operations – fetching instructions, reading operands from memory, and writing results to memory or I/O ports.

- **Instruction Prefetch Queue:** 6 bytes (8086) / 4 bytes (8088). Stores pre-fetched instructions ahead of the EU.
- **Segment Registers (4):** CS (Code Segment), DS (Data Segment), SS (Stack Segment), ES (Extra Segment). Each 16-bit.
- **Instruction Pointer (IP):** 16-bit register holding the offset of the next instruction within the code segment.
- **Address Adder:** Combines segment register value (shifted left 4 bits) with offset to produce 20-bit physical address.
- **Internal Communication Bus:** Connects BIU with EU for passing instructions and data.

Physical Address Formula: Physical Address = (Segment Register × 16) + Offset Register
Example: CS = 1000H, IP = 0200H → Physical Address = 10000H + 0200H = 10200H

5.3 Execution Unit (EU)

The EU receives instructions from the BIU's instruction queue, decodes them, and executes them. It never directly interacts with the system bus.

- **ALU (Arithmetic Logic Unit):** Performs all arithmetic (+, -, ×, ÷) and logic (AND, OR, NOT, XOR) operations.
- **General Purpose Registers:** AX, BX, CX, DX – each 16-bit, can be split into high (H) and low (L) 8-bit halves.

- **Pointer and Index Registers:** SP (Stack Pointer), BP (Base Pointer), SI (Source Index), DI (Destination Index).
- **Flag Register:** 16-bit register with status, control, and system flags.
- **Control Unit:** Decodes instructions and generates control signals for execution.

5.4 Register Organization

The 8086 has **14 registers**, all 16-bit:

5.4.1 General Purpose Registers

Register	Full Name	High Byte	Low Byte	Special Use
AX	Accumulator	AH	AL	Arithmetic, I/O operations
BX	Base Register	BH	BL	Base address for memory addressing
CX	Count Register	CH	CL	Loop counter, shift/rotate count
DX	Data Register	DH	DL	I/O port address, multiply/divide

5.4.2 Segment Registers

Register	Full Name	Points To
CS	Code Segment	Memory segment containing program instructions
DS	Data Segment	Memory segment containing data variables
SS	Stack Segment	Memory segment used for stack operations
ES	Extra Segment	Additional data segment for string operations

5.4.3 Pointer and Index Registers

Register	Full Name	Function
SP	Stack Pointer	Points to top of stack in Stack Segment
BP	Base Pointer	Used to access parameters on the stack
SI	Source Index	Points to source string in DS for string operations
DI	Destination Index	Points to destination string in ES for string operations
IP	Instruction Pointer	Holds offset of next instruction to be executed (in CS)

5.4.4 Flag Register

The **Flag Register** (also called the Status Register) is a 16-bit register. Only 9 of the 16 bits are used. Flags are set/cleared based on the result of operations.

Flag	Symbol	Type	Meaning When SET (=1)
Carry Flag	CF	Status	Carry/borrow from MSB in arithmetic operation

Parity Flag	PF	Status	Result has even number of 1-bits
Auxiliary Carry	AF	Status	Carry from bit 3 to bit 4 (BCD arithmetic)
Zero Flag	ZF	Status	Result of operation is zero
Sign Flag	SF	Status	Result is negative (MSB = 1)
Overflow Flag	OF	Status	Signed arithmetic overflow occurred
Trap Flag	TF	Control	Single-step mode for debugging
Interrupt Flag	IF	Control	Maskable interrupts are enabled
Direction Flag	DF	Control	String operations go from high to low address

5.5 Memory Segmentation

The 8086 uses a **segmented memory model** to address 1 MB of memory using 16-bit registers. The 1 MB memory space is divided into segments of up to 64 KB each.

Each memory location has a **logical address** expressed as Segment:Offset. The physical address is computed as:

Physical Address = (Segment Register × 10H) + Offset

Since multiplying by 10H (16 in decimal) is equivalent to shifting left by 4 bits:
A 16-bit segment value shifted left 4 bits gives a 20-bit base address (range 00000H–FFFF0H).
Adding the 16-bit offset gives a final 20-bit physical address (range 00000H–FFFFFH = 1 MB).

ADDRESS	SEGMENT	REGISTER : PURPOSE
FFFFFH (Top)	**Stack Segment**	SS : Stack operations (PUSH/POP)
	Extra Segment	ES : Extra data / string destination
	Data Segment	DS : Variables and data storage
00000H (Base)	**Code Segment**	CS : Program instructions (code)

CHAPTER 6

Buses in Microprocessor Systems

6.1 Introduction to Bus System

A **bus** is a collection of wires (signal lines) that carry information between the microprocessor, memory, and I/O devices. Think of a bus as a highway: multiple vehicles (data signals) travel together on defined lanes (wires) in an organized manner.

5.2 Why Are Buses Needed?

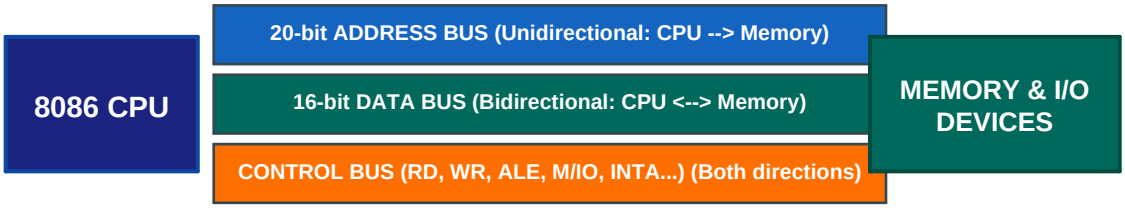
- To provide a **standardized communication path** between CPU, memory, and peripherals
- To reduce the number of connections needed (shared wires instead of dedicated point-to-point wires)
- To allow multiple devices to share the same communication medium
- To simplify system design and reduce hardware cost
- To enable expandability – new devices can be added to the bus

6.3 Types of Buses

Bus Type	Direction	Width (8086)	Purpose
Address Bus	Unidirectional (CPU → Memory/IO)	20 bits	Specifies memory/IO location
Data Bus	Bidirectional (CPU ↔ Memory/IO)	16 bits (8086) / 8 bits (8088)	Transfers data
Control Bus	Bidirectional	Variable	Coordinates bus operations

6.4 Base Bus Concept

The term **Base Bus** refers to the fundamental bus architecture upon which a microprocessor-based system is built. In the 8086 context, the base bus consists of the combination of the Address Bus, Data Bus, and Control Bus working together. All communication in the system flows through this base bus structure.



6.5 Bus Communication Process

A bus cycle describes one complete transfer of data on the bus. A typical 8086 bus cycle:

State	Name	Activity
T1	Address State	CPU drives 20-bit address onto Address Bus; ALE (Address Latch Enable) goes HIGH

T2	Data Setup	CPU floats address bus; data/control lines prepared
T3	Data Transfer	Data is placed on or read from the Data Bus
T4	Bus Free	Bus cycle ends; data latched; bus released

CHAPTER 7

Data Bus

9.1 Definition

The **Data Bus** is a group of parallel signal lines that carry actual data (instructions, operands, results) between the CPU, memory, and I/O devices. It is **bidirectional** – data can flow in both directions (CPU to memory, or memory to CPU) depending on whether a read or write operation is occurring.

> **8086 Data Bus:** 16 bits wide (lines D0–D15) → transfers 2 bytes simultaneously
> **8088 Data Bus:** 8 bits wide (lines D0–D7) → transfers 1 byte at a time

7.2 Functions of Data Bus

- Transfers **instructions** from memory to the CPU for execution
- Carries **data/operands** from memory to CPU for processing
- Carries **results** from CPU back to memory or I/O ports
- Transfers data between **CPU and I/O devices**
- Enables data exchange between **registers and memory**

7.3 Data Transfer Mechanism

During a **memory read**: The CPU first places the memory address on the Address Bus. The memory device decodes the address and places the data byte/word onto the Data Bus. The CPU then reads the data from the Data Bus.

During a **memory write**: The CPU places the address on the Address Bus and the data to be written on the Data Bus. The memory device latches the data.

7.4 Data Bus Width and Impact

Processor	Data Bus Width	Bytes/Transfer	16-bit Transfer Cycles	Performance
8085	8 bits	1 byte	2 cycles	Baseline
8088	8 bits (external)	1 byte	2 cycles	Similar to 8085 on bus
8086	16 bits	2 bytes	1 cycle	~2× faster data transfer
80286	16 bits	2 bytes	1 cycle	Faster clock, better
80386	32 bits	4 bytes	1 cycle	~2× 80286 data width
Pentium	64 bits	8 bytes	1 cycle	Very high bandwidth

7.5 Data Flow Examples

Example 1: MOV AX, [1000H] – Read 16-bit word from memory address 1000H into AX

1	**Place Address**	CPU drives 20-bit address on A0-A19; ALE goes HIGH

2	**Assert RD/WR**	RD goes LOW (read) or WR goes LOW (write); M/IO set
3	**Data Transfer**	Memory drives data on D0-D15 (read) OR CPU drives data (write)
4	**Latch & End**	CPU samples data bus; RD/WR returns HIGH; bus released

CHAPTER 8

Address Bus

9.1 Definition

The **Address Bus** is a group of signal lines that carry the **memory address** or **I/O port address** from the CPU to memory or I/O devices. It is **unidirectional** – addresses flow only from the CPU outward (CPU is always the bus master for addressing). In the 8086, the Address Bus is **20 bits wide** (lines A0–A19).

8.2 Functions of Address Bus

- Identifies the specific **memory location** to be accessed (read or written)
- Identifies the specific **I/O port** for input/output operations
- Selects which **device** on the bus should respond to the current transaction
- Combined with control signals, determines whether memory or I/O is being addressed
- In 8086: lower 16 bits (A0–A15) are used for I/O addressing (64K ports)

8.3 Address Generation Process in 8086

The 8086 generates a 20-bit physical address by combining a **16-bit segment register** value with a **16-bit offset** value inside the BIU's address adder:

Segment Register (CS/DS/SS/ES)	**e.g. 2000H (16-bit)**
Shift Left 4 bits (x 16 = x 10H)	**= 20000H (20-bit base address)**
Add Offset (IP / BX / SP etc.)	**+ 0150H (16-bit offset)**
Physical Address (on Address Bus)	**= 20150H (final 20-bit address)**

8.4 Memory Addressing Capacity

Processor	Address Bus Width	Max. Memory	Formula
8085	16 bits	64 KB	2^16 = 65,536 bytes
8086 / 8088	20 bits	1 MB	2^20 = 1,048,576 bytes
80286	24 bits	16 MB	2^24 = 16,777,216 bytes
80386	32 bits	4 GB	2^32 = 4,294,967,296 bytes
Pentium	32 bits	4 GB	2^32 = 4,294,967,296 bytes (64 GB with PAE)

8.5 Address Calculation Examples

Example 1: DS = 3000H, BX = 0050H → Physical Address = ?

DS × 10H + BX = 30000H + 0050H = **30050H**

Example 2: SS = 4000H, SP = 0200H → Physical Address = ?

SS × 10H + SP = 40000H + 0200H = **40200H**

CHAPTER 9

Control Bus

9.1 Definition

The **Control Bus** is a collection of signal lines that carry control and timing signals between the CPU, memory, and I/O devices. Unlike the Address and Data buses which are dedicated groups of lines, the Control Bus consists of **individual signal lines**, each with a specific purpose. Some are driven by the CPU (output), and some are received by the CPU (input).

9.2 Key Control Signals in 8086 (Minimum Mode)

Signal	Direction	Function
RD (Read)	Output (CPU→Memory)	LOW = CPU is reading data from memory or I/O
WR (Write)	Output (CPU→Memory)	LOW = CPU is writing data to memory or I/O
ALE (Address Latch Enable)	Output	HIGH pulse indicates valid address on bus
M/IO	Output	HIGH = memory access; LOW = I/O port access
DT/R (Data Transmit/Receive)	Output	Controls direction of data bus transceiver
DEN (Data Enable)	Output	Enables data bus transceiver
INTA (Interrupt Acknowledge)	Output	CPU acknowledging an interrupt request
INTR (Interrupt Request)	Input (→CPU)	External device requests a maskable interrupt
NMI (Non-Maskable Interrupt)	Input (→CPU)	High-priority interrupt that cannot be disabled
RESET	Input (→CPU)	Resets the CPU to initial state
READY	Input (→CPU)	Memory/IO signals it is ready for transfer
CLK (Clock)	Input (→CPU)	System clock signal for synchronization
HOLD / HLDA	Bidirectional	Bus arbitration – another master requests/gets bus

9.3 Read Operation (Memory Read Cycle)

T1	T2	T3	T4
Address on Bus ALE = HIGH	Data Bus Prepared RD/WR = LOW	Data Transfer on Data Bus	Bus Cycle Ends Data Latched

9.4 Write Operation (Memory Write Cycle)

1	**Place Address**	CPU drives 20-bit address on A0-A19; ALE goes HIGH
2	**Assert RD/WR**	RD goes LOW (read) or WR goes LOW (write); M/IO set
3	**Data Transfer**	Memory drives data on D0-D15 (read) OR CPU drives data (write)
4	**Latch & End**	CPU samples data bus; RD/WR returns HIGH; bus released

9.5 Interrupt Signals

• **INTR (Interrupt Request):** Maskable interrupt – can be enabled/disabled using IF flag. When INTR is asserted and IF=1, CPU completes current instruction then responds.

• **NMI (Non-Maskable Interrupt):** Cannot be disabled by software. Always responded to. Used for critical events like power failure.

• **INTA (Interrupt Acknowledge):** CPU sends two INTA pulses to the interrupting device; the device then puts the interrupt type number on the data bus.

CHAPTER 10

Introduction to 80286 Microprocessor

12.1 Historical Background

The **Intel 80286** (commonly called 286) was introduced in **February 1982**. It was the successor to the 8086 and was used in the **IBM PC/AT (1984)**. The 80286 introduced the crucial concept of **Protected Mode**, which allowed multitasking operating systems and memory protection.

10.2 Features of 80286

- **16-bit data bus** (same as 8086)
- **24-bit address bus** → up to **16 MB** of physical memory (vs 1 MB for 8086)
- Clock speeds: **6 MHz, 8 MHz, 10 MHz, 12.5 MHz**
- Approximately **3–6 times faster** than equivalent 8086 clock speed
- Introduced **Protected Mode** with memory protection and multitasking
- In protected mode, can address **1 GB of virtual memory** per task
- Has a built-in **Memory Management Unit (MMU)**
- Contains **134,000 transistors**
- Fully backward-compatible with 8086/8088 software (in Real Mode)
- Available in **68-pin PLCC** and **68-pin LCC** packages

10.3 Operating Modes

Real Mode:

The 80286 starts in Real Mode on power-up, acting exactly like a fast 8086. It uses the same segmented 20-bit addressing, can access only 1 MB of memory, and runs all 8086/8088 software without modification.

Protected Mode:

Protected Mode uses **24-bit physical addresses** to access up to 16 MB of RAM. Each task gets its own memory space. The processor enforces access rights – a program cannot access memory belonging to another program or the OS kernel without proper privilege.

Feature	Real Mode	Protected Mode
Addressing	20-bit (1 MB)	24-bit (16 MB)
Memory Protection	None	Full hardware protection
Multitasking	No	Yes (hardware supported)
Virtual Memory	No	1 GB per task
8086 Compatibility	Full	Partial (V86 mode needed for 8086 apps)
Privilege Levels	None	4 levels (Ring 0–3)

10.4 Advantages of 80286 over 8086

- 24-bit address bus → 16× more addressable memory (16 MB vs 1 MB)
- Protected Mode enables true multitasking and memory protection
- Faster execution: improved microcode and higher clock speeds
- Built-in MMU for memory segmentation in protected mode
- Hardware support for privilege levels (OS protection)

CHAPTER 11

Introduction to 80386 Microprocessor

12.1 Historical Background

The **Intel 80386** (also called i386 or 386) was introduced in **October 1985**. It was the first **fully 32-bit** x86 processor – with a 32-bit internal architecture, 32-bit data bus, and 32-bit address bus. The 386 was a major milestone in PC history and enabled the success of UNIX and early versions of Windows.

11.2 Features of 80386

- **32-bit internal architecture** (all registers 32-bit)
- **32-bit data bus** – transfers 4 bytes in one cycle
- **32-bit address bus** – addresses up to **4 GB** of physical memory
- Clock speeds: **16 MHz, 20 MHz, 25 MHz, 33 MHz**
- Contains approximately **275,000 transistors**
- Three operating modes: **Real Mode, Protected Mode, Virtual 8086 Mode**
- Built-in **Memory Management Unit (MMU)** with segmentation and paging
- Supports **virtual memory** up to **64 TB** per task (with paging)
- Hardware support for **paging** – 4 KB page size
- Full backward-compatibility with 8086 and 80286 software
- Available in **132-pin PGA** (Pin Grid Array) package
- Also available as 80386SX (16-bit external bus, 24-bit address bus – budget version)

11.3 32-bit Register Set

The 80386 extended all 16-bit registers of the 8086 to 32 bits by adding an 'E' prefix:

8086 Register (16-bit)	80386 Register (32-bit)	Notes
AX	EAX (bits 0–31, AX = bits 0–15)	Extended Accumulator
BX	EBX	Extended Base
CX	ECX	Extended Count
DX	EDX	Extended Data
SP	ESP	Extended Stack Pointer
BP	EBP	Extended Base Pointer
SI	ESI	Extended Source Index
DI	EDI	Extended Destination Index
IP	EIP	Extended Instruction Pointer
FLAGS	EFLAGS	Extended Flags (more flag bits added)

11.4 Operating Modes of 80386

- **Real Mode:** Same as 8086 real mode. CPU starts here on power-up. Limited to 1 MB memory.
- **Protected Mode:** Full 32-bit addressing (4 GB), memory protection, multitasking, privilege levels.
- **Virtual 8086 Mode (V86):** Runs multiple 8086 programs simultaneously as separate 'virtual machines' within Protected Mode. Allows DOS programs to run inside Windows.

11.5 Paging Concept

Paging is a memory management technique introduced in the 80386 that allows the operating system to use **non-contiguous physical memory** as if it were contiguous. Memory is divided into fixed-size blocks called **pages** (4 KB each). A **Page Table** maps virtual page numbers to physical page frame numbers.

Paging Formula:
Virtual Address → Page Directory → Page Table → Physical Frame
80386: 32-bit virtual address = 10-bit directory + 10-bit page table + 12-bit offset
Each page = 4 KB = 4096 bytes

CHAPTER 12

Introduction to Pentium Chips

12.1 Historical Background & Evolution

Intel's original plan was to call the successor to the 80486 the **'80586'**, but since numbers cannot be trademarked, Intel named it **Pentium** (from Greek 'pente' = five). The **original Pentium (P5)** was introduced in **March 1993** at 60 MHz and 66 MHz. It represented a **quantum leap** in PC performance.

12.2 Key Features of Pentium

- **32-bit internal architecture** (like 80386/80486) – all registers 32-bit
- **64-bit external data bus** – transfers 8 bytes in one cycle (double the 80486)
- **32-bit address bus** – addresses up to 4 GB physical memory
- **Superscalar architecture** – two parallel integer pipelines (U-pipe and V-pipe)
- **Clock speeds:** 60, 66, 75, 90, 100, 120, 133, 150, 166, 200 MHz
- Contains approximately **3.1 million transistors** (P5 original)
- Separate **8 KB instruction cache** and **8 KB data cache** (L1 cache)
- Built-in **Floating-Point Unit (FPU)** – much faster than 80486 FPU
- **5-stage integer pipeline** per pipe (PF, D1, D2, EX, WB)
- Supports **System Management Mode (SMM)** for power management
- Available in **273-pin PGA** package (socket 4/5)
- Manufactured in **0.8 µm BICMOS technology** (later 0.6 µm, 0.35 µm)

12.3 Superscalar Architecture

The Pentium's most revolutionary feature is its **superscalar architecture**: two independent integer execution pipelines called the **U-pipe** and **V-pipe** that can execute **two instructions simultaneously** in the same clock cycle.

	PREFETCH (PF)	DECODE 1 (D1)	DECODE 2 (D2)	EXECUTE (EX)	WRITE BACK (WB)
U-PIPE (Main)	ACTIVE	ACTIVE	ACTIVE	ACTIVE	ACTIVE
V-PIPE (Secondary)	ACTIVE	ACTIVE	ACTIVE	ACTIVE	ACTIVE

12.4 Pipeline Processing

Each Pentium pipeline has **5 stages**:

Stage	Abbreviation	Function
Prefetch	PF	Fetches instructions from cache into prefetch buffer
Instruction Decode 1	D1	Decodes instruction opcode
Instruction Decode 2	D2	Calculates memory addresses if needed

Execute	EX	Performs the operation (ALU, FPU, etc.)
Write Back	WB	Writes result to register or cache

12.5 Cache Memory

The Pentium has **two separate on-chip L1 caches** (Harvard cache architecture):

- **8 KB Instruction Cache:** Stores recently fetched instructions. Read-only. Reduces memory fetch latency.
- **8 KB Data Cache:** Stores recently read/written data. Read-write. Write-back or write-through policy.
- **L2 Cache:** Not on-chip for original Pentium – typically 256 KB or 512 KB external L2 cache on motherboard.

Why Separate Caches? Separating instruction and data caches prevents structural hazards – both pipelines can simultaneously access their respective caches without conflicts (one fetching instructions, the other reading/writing data).

CHAPTER 13

Comparative Study of Microprocessors

13.1 Master Comparison Table

Feature	8086	8088	80286	80386	Pentium
Year	1978	1979	1982	1985	1993
Data Bus (ext)	16-bit	8-bit	16-bit	32-bit	64-bit
Address Bus	20-bit	20-bit	24-bit	32-bit	32-bit
Max Memory	1 MB	1 MB	16 MB	4 GB	4 GB
Registers	16-bit	16-bit	16-bit	32-bit	32-bit
Max Clock	10 MHz	8 MHz	12.5 MHz	33 MHz	200 MHz+
Transistors	~29K	~29K	~134K	~275K	~3.1M
Instr. Queue	6 bytes	4 bytes	6 bytes	16 bytes	32 bytes
Pipelines	1 (2-stage)	1 (2-stage)	1 (6-stage)	1 (6-stage)	2 (5-stage each)
Virtual Memory	No	No	1 GB/task	64 TB/task	64 TB/task
Paging	No	No	No	Yes (4KB)	Yes (4KB)
FPU	External (8087)	External (8087)	External (80287)	External (80387)	On-chip
L1 Cache	None	None	None	None	16 KB (split)
Technology	HMOS	HMOS	HMOS III	CMOS	BiCMOS
Package	40-DIP	40-DIP	68-PLCC	132-PGA	273-PGA

13.2 Performance Evolution Timeline

1978	**8086**	16-bit ext. bus, 1 MB RAM, 5-10 MHz
1979	**8088**	8-bit ext. bus, IBM PC, 5-8 MHz
1982	**80286**	Protected Mode, 16 MB, 6-12.5 MHz
1985	**80386**	Full 32-bit, 4 GB, Paging, V86, 16-33 MHz
1989	**80486**	On-chip FPU + Cache, 25-100 MHz [ref]
1993	**Pentium**	Superscalar dual pipeline, 64-bit bus, 200 MHz

13.3 8086 vs 8088 (Quick Reference)

Aspect	8086	8088
External Data Bus	16 bits (faster)	8 bits (slower, cheaper)
Prefetch Queue	6 bytes	4 bytes
16-bit Transfer	1 bus cycle	2 bus cycles
Cost	More expensive	Less expensive
Famous Use	Embedded/industrial	IBM PC (1981)

— End of Study Guide —